📦 Unit 9: Volume of 3D Solids

Name _____ Date _____ Period _____

Cylinders · Cones · Spheres · Composite Solids · NC.8.G.9

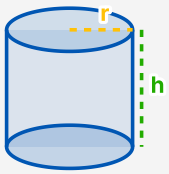
🎯 By the end of this unit, I can…

- Find the volume of cylinders, cones, and spheres
- Explain why a cone is one-third of its matching cylinder
- Solve real-world and composite-solid volume problems

Key Formulas

🔑 **All volume formulas use π (pi ≈ 3.14).** On the EOG you can leave answers in terms of π (like 250π) **or** use 3.14. These two methods give answers that differ by a little — both are correct. The decimals below use 3.14. Always include **cubic units** (cm³, in³, ft³, etc.).

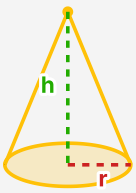
🥫 Cylinder



$$V = \pi r^2 h$$

Base area × height

🍦 Cone



$$V = \frac{1}{3}\pi r^2 h$$

One-third of a cylinder!

🏀 Sphere



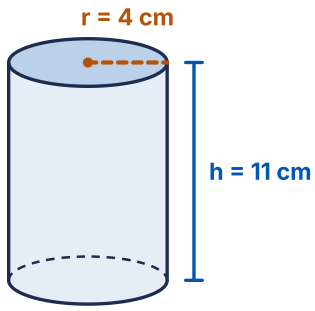
$$V = \frac{4}{3}\pi r^3$$

Only needs the radius!

✏ Practice: Pick the Formula

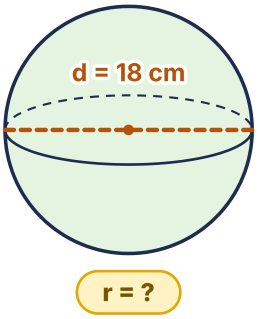
1. ★ Try it This soup can is a cylinder. Which formula finds its volume?

- ○ $V = \pi r^2 h$
- ○ $V = \frac{1}{3}\pi r^2 h$
- ○ $V = \frac{4}{3}\pi r^3$



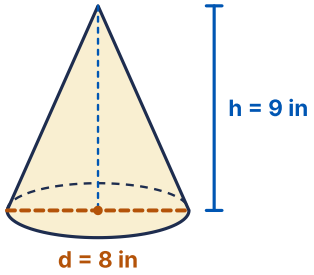
2. ★★ On level This ball has a diameter of 18 cm. What radius goes into $V = \frac{4}{3}\pi r^3$? Type just the number.

Answer: _____



3. ★★★ Challenge This cone has a diameter of 8 in and a height of 9 in. Which setup is correct?

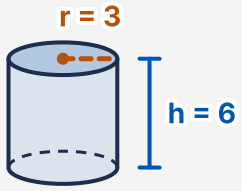
- ○ $V = \frac{1}{3}\pi(4)^2(9)$
- ○ $V = \pi(4)^2(9)$
- ○ $V = \frac{1}{3}\pi(8)^2(9)$
- ○ $V = \frac{1}{3}\pi(9)^2(4)$



Try It: Change r and h

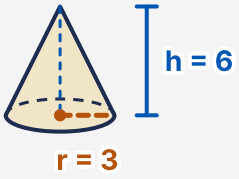
Here r = 3 and h = 6 for all three solids. Compare their volumes. Notice: the cone is always ⅓ of the cylinder. The sphere only cares about **r**.

radius r = **3** height h = **6**



Cylinder

$\pi(3)^2(6) = 54\pi \approx 169.6$



Cone

$\frac{1}{3}\pi(3)^2(6) = 18\pi \approx 56.5$



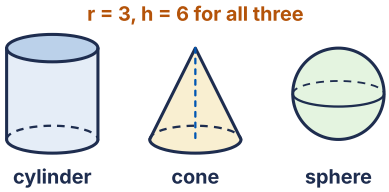
Sphere (uses r only)

$\frac{4}{3}\pi(3)^3 = 36\pi \approx 113.0$

✏ Practice: What Changes the Volume?

1. ★ Try it Which solid's volume formula does NOT use the height h?

- ○ Cylinder
- ○ Cone
- ○ Sphere

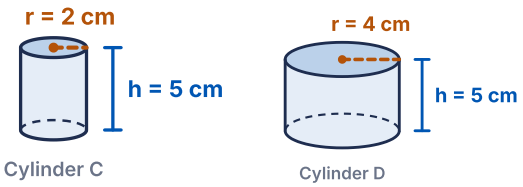


2. ★★ On level Cylinder B is twice as tall as cylinder A. Same radius. How many times bigger is B's volume? Type just the number.

Answer: _____



3. ★★★ Challenge Cylinder D has double the radius of cylinder C. Same height. How many times bigger is D's volume? Type just the number.

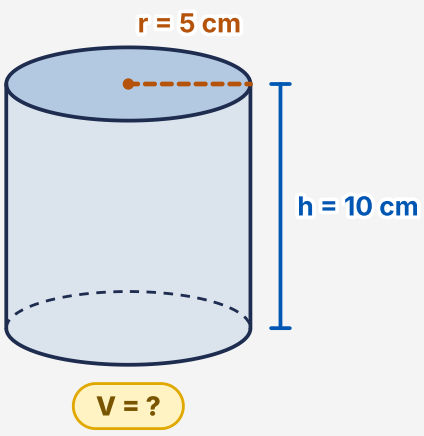


Answer: _____

Worked Examples: 🥫 Cylinders

$V = \pi r^2 h$. Square the radius, multiply by π, then multiply by the height.

🥫 Cylinder: r = 5 cm, h = 10 cm



$V = \pi r^2 h$

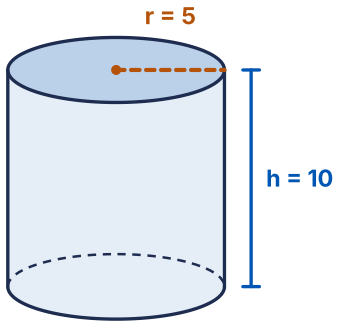
$V = \pi(5)^2(10)$

$V = \pi(25)(10)$

$V = 250\pi$

$V \approx 785.0\ cm^3$ (using 3.14; a calculator gives ≈ 785.4)

🔍 Spot the Error: Cylinder: r = 5, h = 10



× Student: V = π(5)(10) = 50π ≈ 157.1

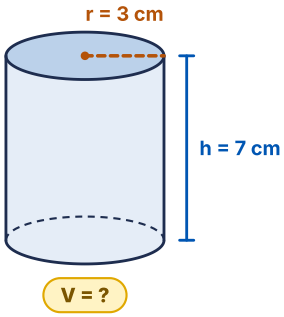
✓ Fix: Forgot to SQUARE the radius! V = π(5²)(10) = π(25)(10) = 250π ≈ 785.4

💡 The formula is πr²h — that ² on the radius is critical!

✏ Practice: Cylinders

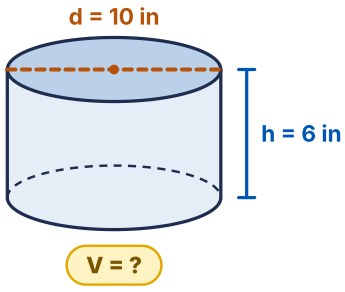
1. ★ Try it Find the volume. Leave your answer in terms of π.

Answer: _____

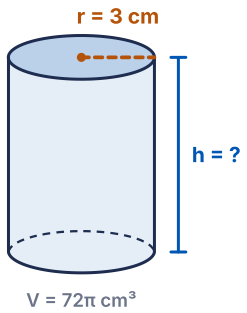


2. ★★ On level Find the volume. Use 3.14 and round to the nearest tenth.

Answer: _____



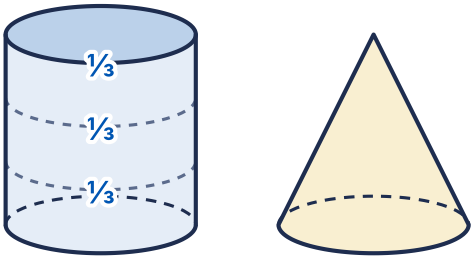
3. ★★★ Challenge This can holds 72π cm³. Its radius is 3 cm. How tall is it? Type just the number.



Answer: _____

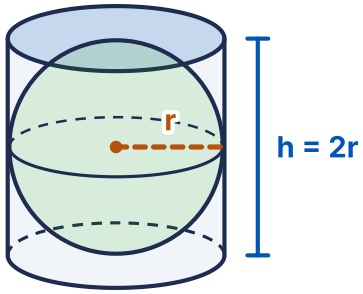
Relationship: Cylinder, Cone & Sphere

🔑 **THE BIG RELATIONSHIP:** A cone is exactly ⅓ of a cylinder with the same radius and height. Two cones = ⅔ cylinder. So a cone's volume is always ⅓ of the matching cylinder.



cylinder **cone (same r and h)**

Fill the cone with water and pour it into the cylinder. It takes **3 cones** to fill it. So one cone = ⅓ of the cylinder.



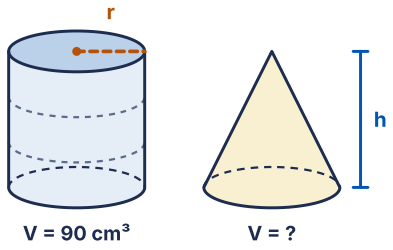
A sphere that just fits inside a cylinder (the height is 2r) fills ⅔ of it.

⚠️ **SPHERE SHORTCUT:** If the sphere fits perfectly inside a cylinder (diameter = height = 2r), the sphere takes up exactly ⅔ of the cylinder's volume. This is why $V = \frac{4}{3}\pi r^3$.

✏ Practice: Cone, Cylinder & Sphere

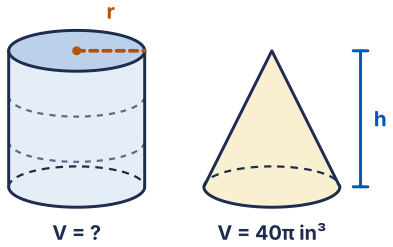
1. ★ Try it The cylinder holds 90 cm³. The cone has the SAME radius and height. What is the cone's volume?

- ○ 30 cm³
- ○ 45 cm³
- ○ 90 cm³
- ○ 270 cm³



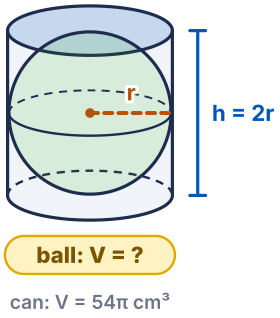
2. ★★ On level The cone holds 40π in³. The cylinder has the same radius and height. Find the cylinder's volume. Leave it in terms of π.

Answer: _____



3. ★★★ Challenge A ball fits exactly inside a can (h = 2r). The can holds 54π cm³. Find the ball's volume. Leave it in terms of π.

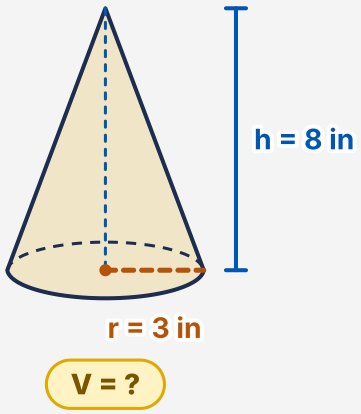
Answer: _____



Worked Examples: 🍦 Cones

$V = \frac{1}{3}\pi r^2 h$. Work it like a cylinder, then take ⅓ (divide by 3).

🍦 Cone: r = 3 in, h = 8 in



$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3} \cdot \pi(3)^2(8)$

$V = \frac{1}{3} \cdot \pi(9)(8)$

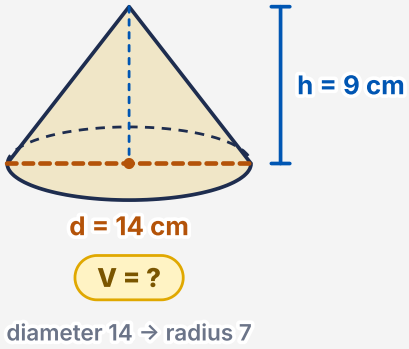
$V = \frac{1}{3} \cdot 72\pi$

$V = 24\pi$

$V \approx 75.4\ in^3$

📝 Worked Example

Volume of a cone: diameter = 14 cm, height = 9 cm



diameter 14 → radius 7

Step 1: Diameter = 14, so r = 7. h = 9

Step 2: Formula: $V = \frac{1}{3}\pi r^2 h$

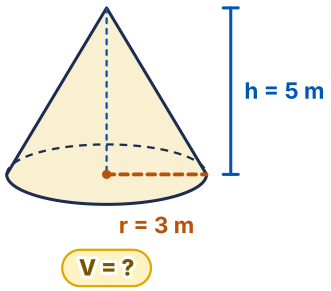
Step 3: Substitute: $V = \frac{1}{3}\pi(7^2)(9) = \frac{1}{3}\pi(49)(9)$

Step 4: Multiply: $V = \frac{1}{3}(441\pi)$

Step 5: $V = 147\pi \approx 461.8\ cm^3$ ✓

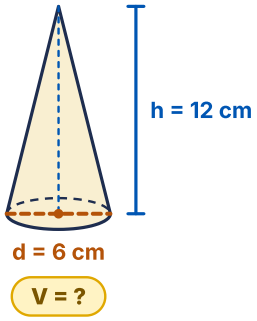
✏ Practice: Cones

1. ★ Try it Find the volume. Leave your answer in terms of π.



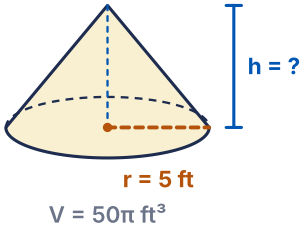
Answer: _____

2. ★★ On level This cone has a diameter of 6 cm. Find its volume. Use 3.14 and round to the nearest tenth.



Answer: _____

3. ★★★ Challenge This cone holds 50π ft³. Its radius is 5 ft. Find its height. Type just the number.

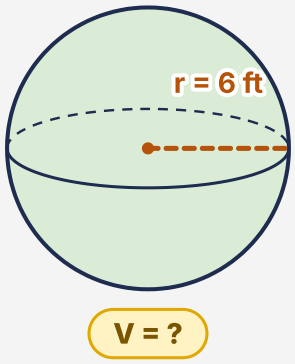


Answer: _____

Worked Examples: 🏀 Spheres

$V = \frac{4}{3}\pi r^3$. CUBE the radius (r × r × r), then multiply by $\frac{4}{3}$ and π.

🏀 Sphere: r = 6 ft



$V = \frac{4}{3}\pi r^3$

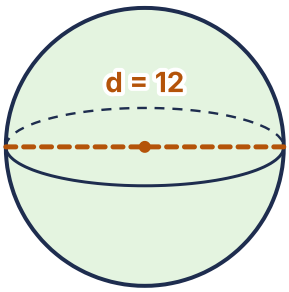
$V = \frac{4}{3} \cdot \pi(6)^3$

$V = \frac{4}{3} \cdot \pi(216)$

$V = \frac{4}{3} \cdot 216\pi = 288\pi$

$V \approx 904.3$ ft³ (using 3.14; a calculator gives ≈ 904.8)

🔍 Spot the Error: Sphere: diameter = 12



the 12 goes all the way across

× Student: r = 12, $V = \frac{4}{3}\pi(12^3) = \frac{4}{3}\pi(1728) \approx 7238.2$

✓ Fix: Used DIAMETER instead of radius! r = 6, not 12. $V = \frac{4}{3}\pi(6^3) = \frac{4}{3}\pi(216) = 288\pi \approx 904.8$

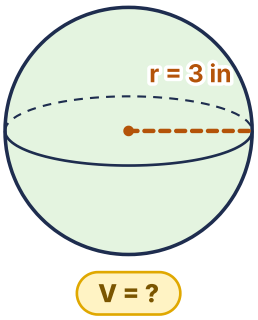
💡 Always check: did the problem say diameter or radius? Diameter ÷ 2 = radius.

🔑 **HEMISPHERE = half a sphere** (like a bowl or a dome). Find the whole sphere, then divide by 2: $V = \frac{1}{2} \cdot \frac{4}{3}\pi r^3$.

✏ Practice: Spheres

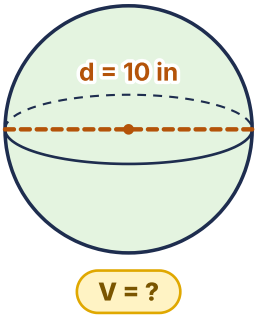
1. ★ Try it Find the volume. Leave your answer in terms of π.

Answer: _____



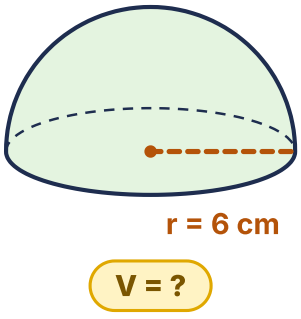
2. ★★ On level A ball has a diameter of 10 in. Find its volume. Use 3.14 and round to the nearest tenth.

Answer: _____



3. ★★★ Challenge This bowl is a hemisphere with radius 6 cm. Find its volume. Leave your answer in terms of π.

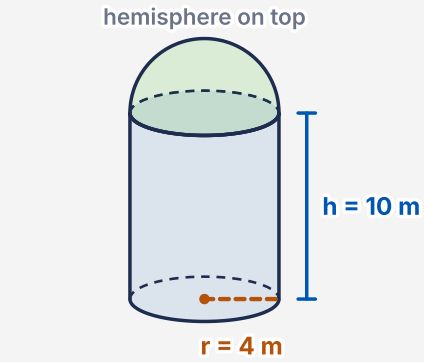
Answer: _____



Worked Examples: 📦 Composite & Real-World

A composite solid is made of simple solids. Find the volume of each part. Then add them.

📦 Composite: Cylinder + Hemisphere (r = 4 m, h = 10 m)



V = cylinder + half sphere

Cylinder: $V = \pi(4)^2(10) = 160\pi$

Full sphere: $V = {}^4/_3\pi(4)^3 = 256\pi/3$

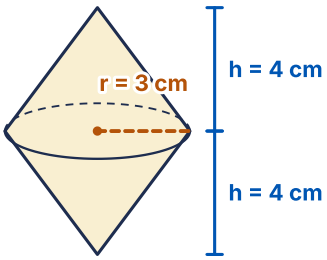
Hemisphere = half: $256\pi/6 = 128\pi/3 \approx 42.67\pi$

Total = $160\pi + 128\pi/3 = 202.67\pi \approx 636.4$ m³

(using 3.14; a calculator gives ≈ 636.7)

✏ Practice: Composite & Real-World

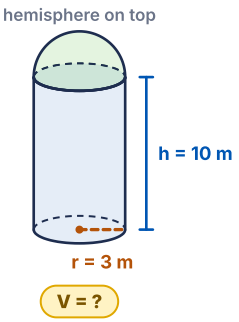
1. ★ Try it Two equal cones are glued base to base. Find the total volume. Leave your answer in terms of π.



V = ?

Answer: _____

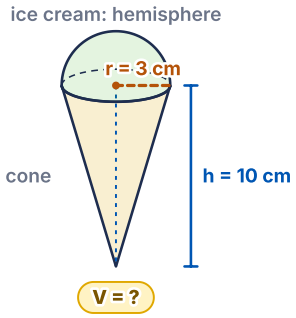
2. ★★ On level A grain silo is a cylinder with a hemisphere on top. Find its volume. Use 3.14 and round to the nearest tenth.



V = ?

Answer: _____

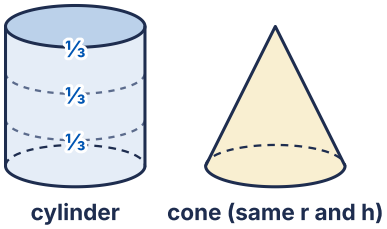
3. ★★★ Challenge An ice cream cone is filled, with a hemisphere of ice cream on top. Find the total volume. Leave your answer in terms of π.



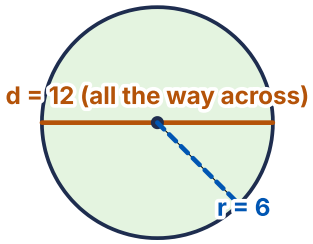
Answer: _____

Common Mistakes

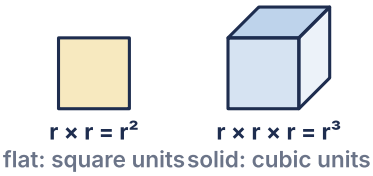
❌ **Forgetting ⅓ for cones** — The cone formula has ⅓. Without it, you get the cylinder's volume!



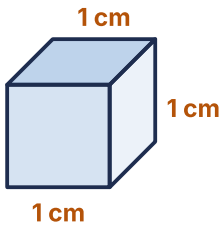
❌ **Using diameter instead of radius** — If the problem gives diameter, divide by 2 first!



❌ **Forgetting to cube for spheres** — The sphere uses r^3, not r^2. Square the radius for cylinders/cones, CUBE it for spheres.



❌ **Missing units** — Volume is always in CUBIC units (cm^3, in^3, ft^3).

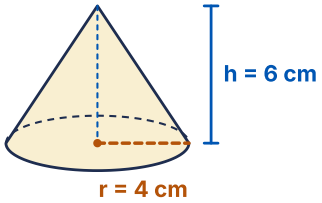


1 cm^3 = one cubic unit

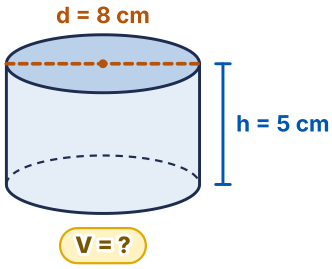
✏ Practice: Spot the Mistake

1. ★ Try it — A student found this cone's volume: $V = \pi(4)^2(6) = 96\pi$. What did they forget?

- ○ the ⅓
- ○ to cube the radius
- ○ to cut the diameter in half

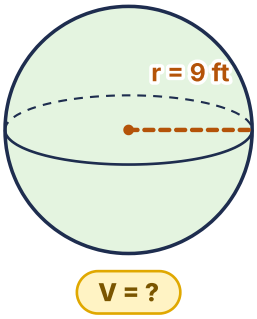


2. ★★ On level — Mia used r = 8 for this cylinder. Find the correct volume. Leave your answer in terms of π.



Answer: _____

3. ★★★ Challenge — Sam wrote $V = \frac{4}{3}\pi(9)^2 = 108\pi$ for this sphere. Find the correct volume. Leave your answer in terms of π.



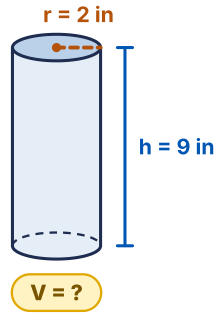
Answer: _____

Putting It All Together

Mixed practice from every section. Read each one: which solid is it? Radius or diameter? 3.14 or π?

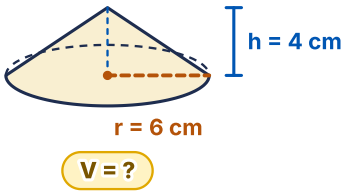
✏ Comprehensive Practice All sections

1. ★ Try it Find the volume of the cylinder. Leave your answer in terms of π.



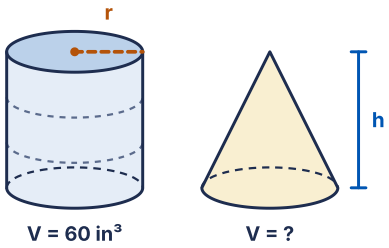
Answer: _____

2. ★ Try it Find the volume of the cone. Leave your answer in terms of π.



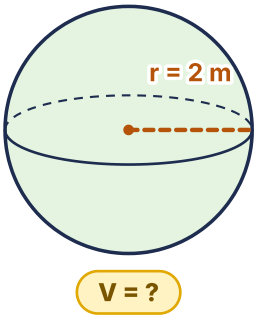
Answer: _____

3. ★ Try it The cylinder holds 60 in³. The cone has the same radius and height. Find the cone's volume. Type just the number.



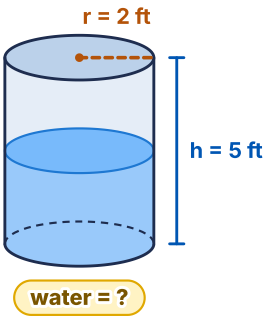
Answer: _____

4. ★★ On level Find the volume of the sphere. Use 3.14 and round to the nearest tenth.



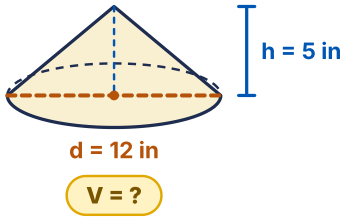
Answer: _____

5. ★★ On level This water tank is a cylinder. It is half full. How much water is in it? Leave your answer in terms of π.



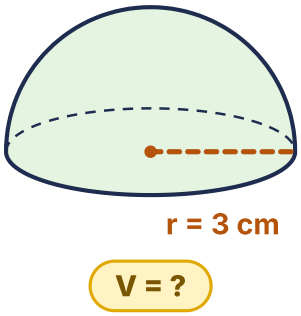
Answer: _____

6. ★★ On level Find the volume of the cone. Use 3.14 and round to the nearest tenth.



Answer: _____

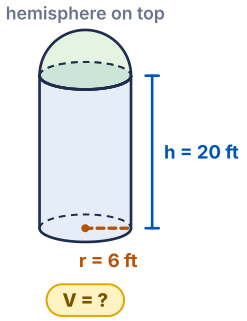
7. ★★ On level Find the volume of the hemisphere. Leave your answer in terms of π.



Answer: _____

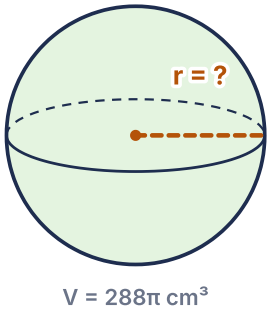
8. ★★★ Challenge A silo is a cylinder with a hemisphere on top. Find its total volume. Leave your answer in terms of π.

Answer: _____



9. ★★★ Challenge A sphere holds 288π cm³. Find its radius. Type just the number.

Answer: _____



10. ★★★ Challenge Which cup holds more: the cylinder cup or the cone cup?

- ○ the cylinder cup
- ○ the cone cup
- ○ they hold the same

